Are LLMs Robust Enough for Operations Research?

An Adversarial Attack and Defense Study via AttackOR

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Why OR Robustness Matters?

Question:
A company needs to allocate products to warehouses. There are several products, such as bread, umbrellas, and glass, with demands of 120, 80, and 100 units, respectively. The company can use several candidate warehouses, including Warehouse 1, Warehouse 2, and Warehouse 3, whose capacities are 150 .... Assigning products to warehouses incurs different costs depending on the product-warehouse pair, which are given as .... Warehouses 2 are typically preferred for handling fragile goods ...
Determine an allocation plan that minimizes the total cost while satisfying demand and capacity constraints.

Misled Minimize total cost ✓

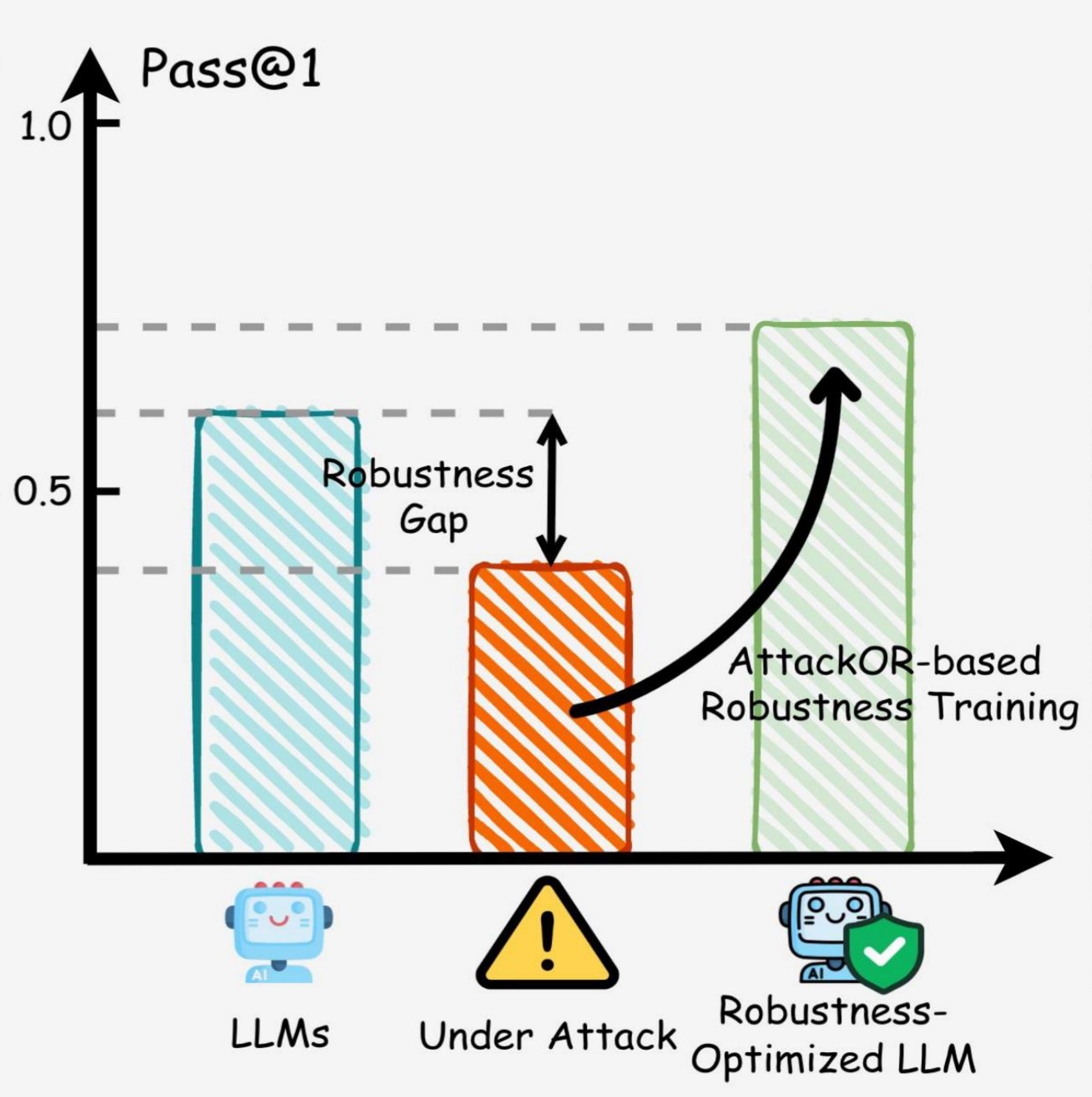
s.t.
- demand for bread, umbrellas, and glass is fully satisfied
- warehouse capacities are respected
- glass is assigned to Warehouse 2 ✗
- all allocation quantities are nonnegative
- ...

Optimal cost = 14,560 ✗

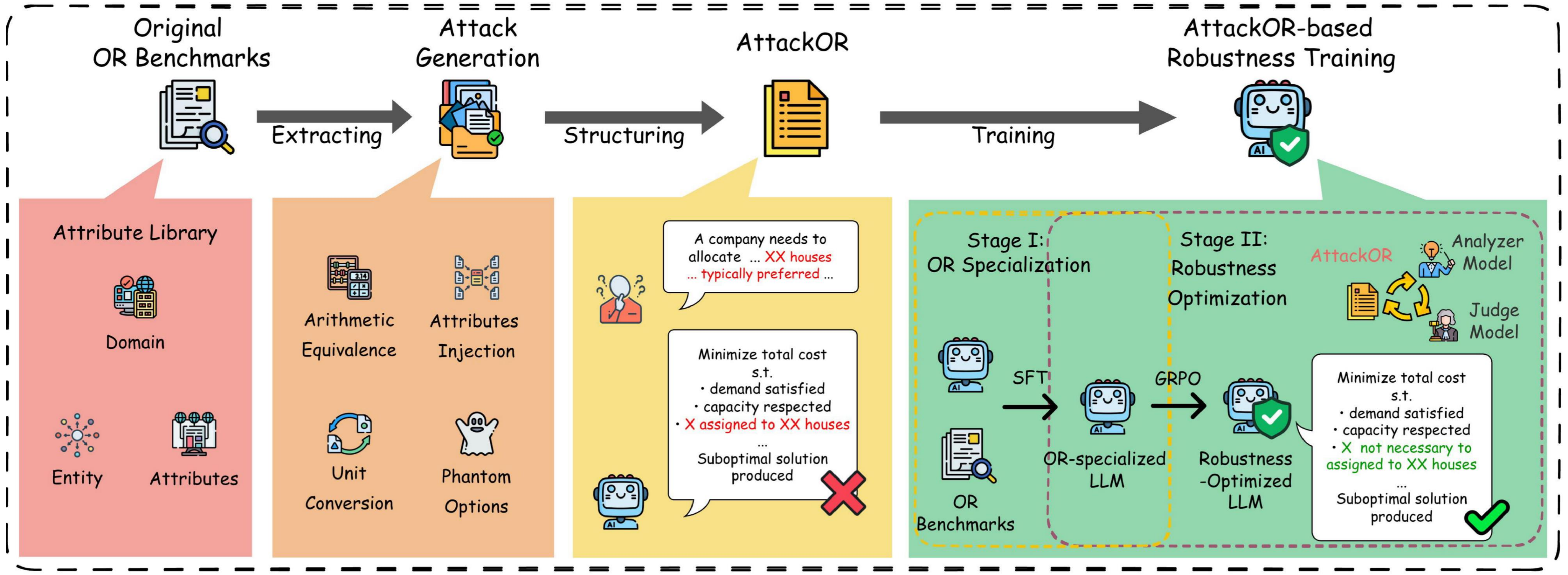
Robust Minimize total cost ✓

s.t.
- demand for bread, umbrellas, and glass is fully satisfied
- warehouse capacities are respected
- glass is not necessarily assigned to Warehouse 2
- all allocation quantities are nonnegative ✓
- ...

Optimal cost = 12,480 ✓



AttackOR Framework



A company needs to allocate ... XX houses ... typically preferred ...

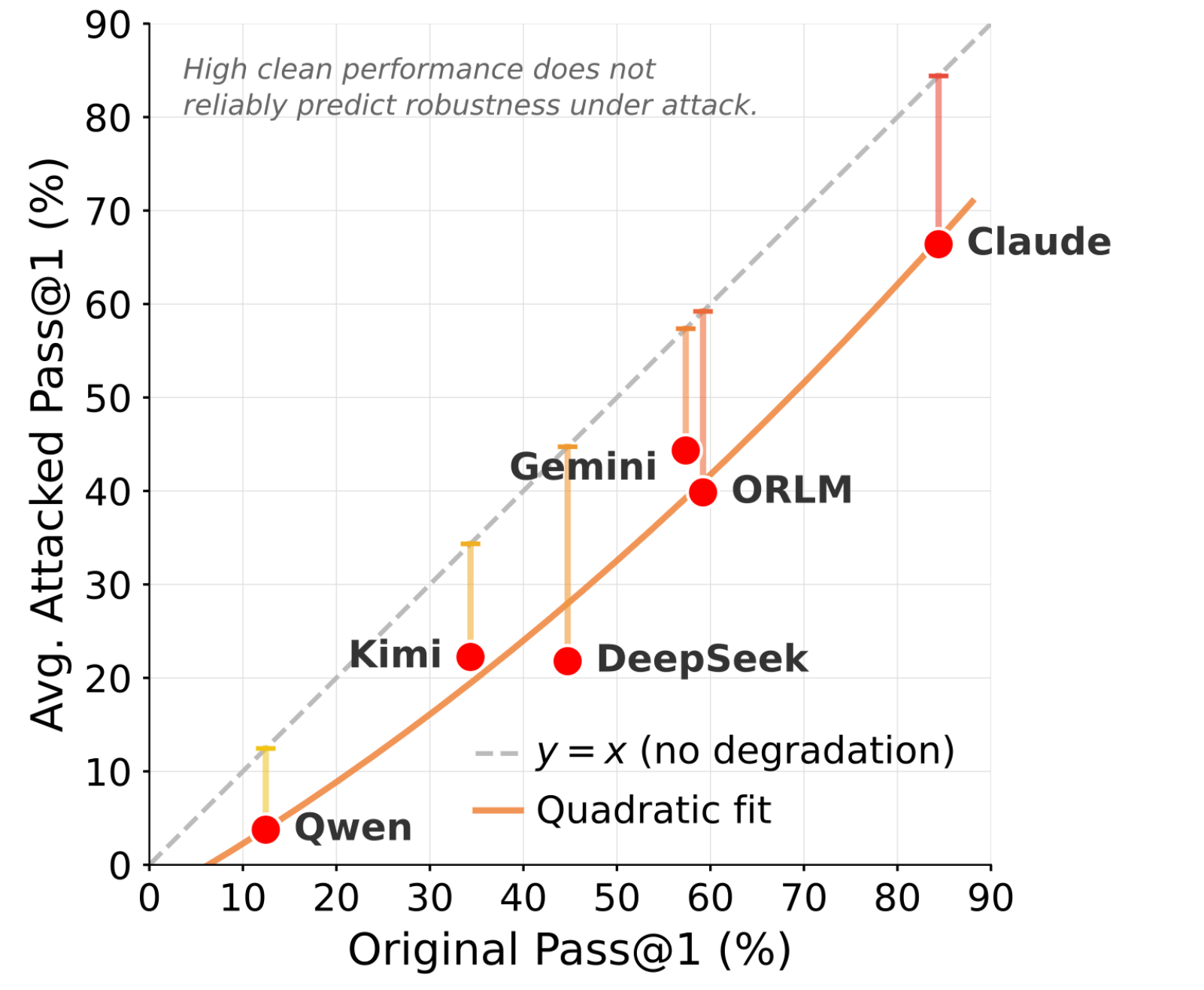
Minimize total cost
s.t.
- demand satisfied
- capacity respected
- X assigned to XX houses
- ...

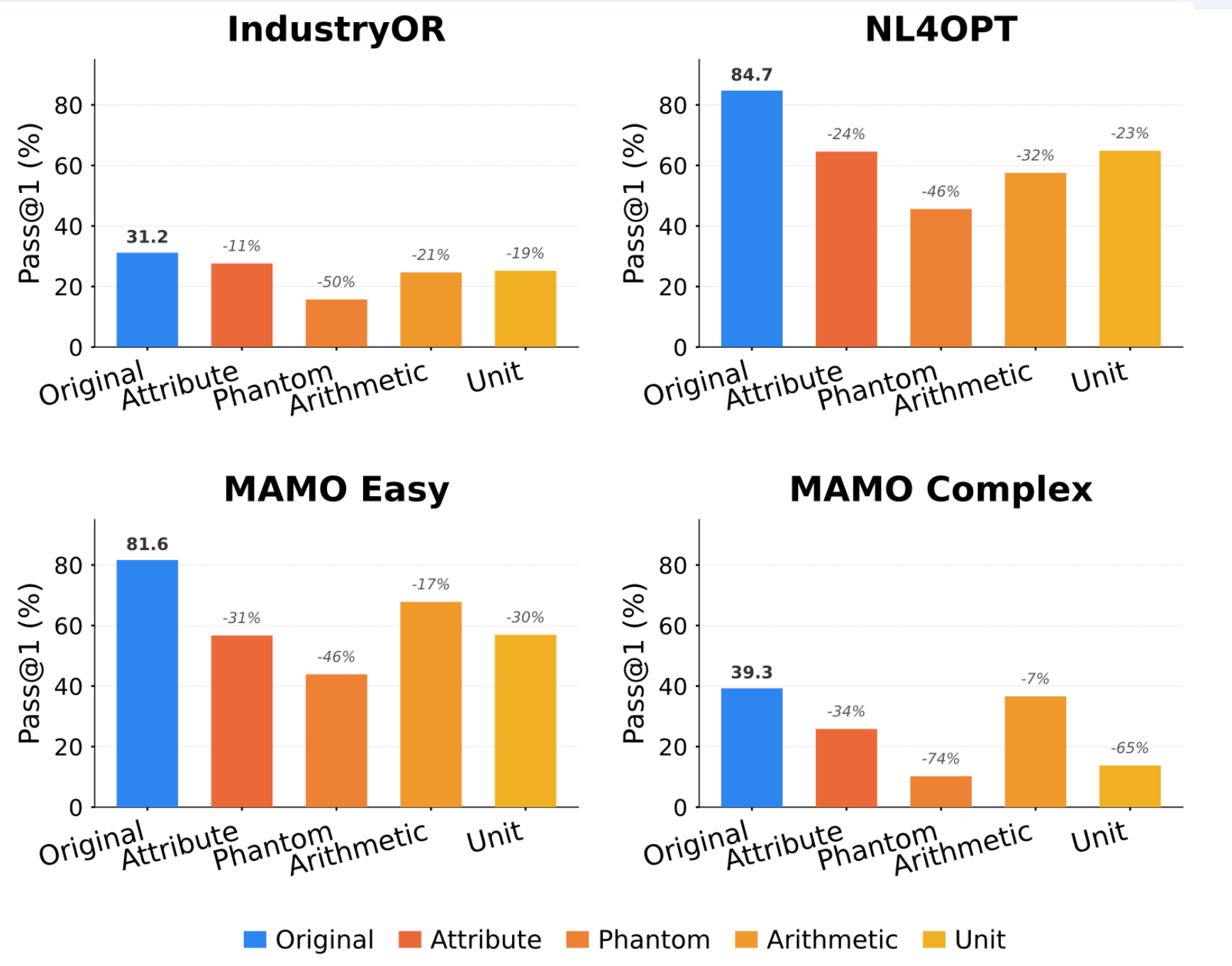
Suboptimal solution produced ✗

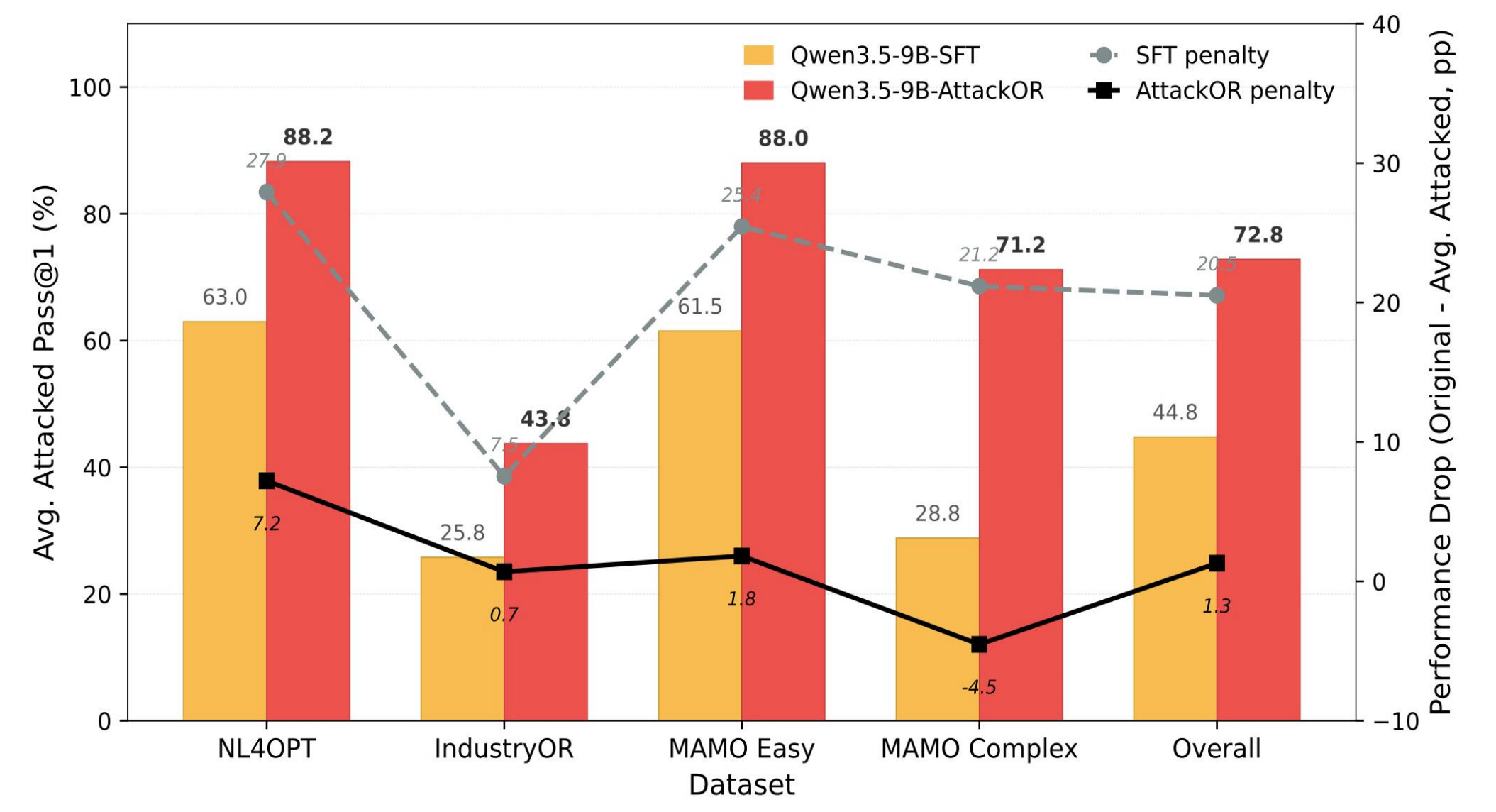
Minimize total cost
s.t.
- demand satisfied
- capacity respected
- X not necessary to assigned to XX houses
- ...

Suboptimal solution produced ✓

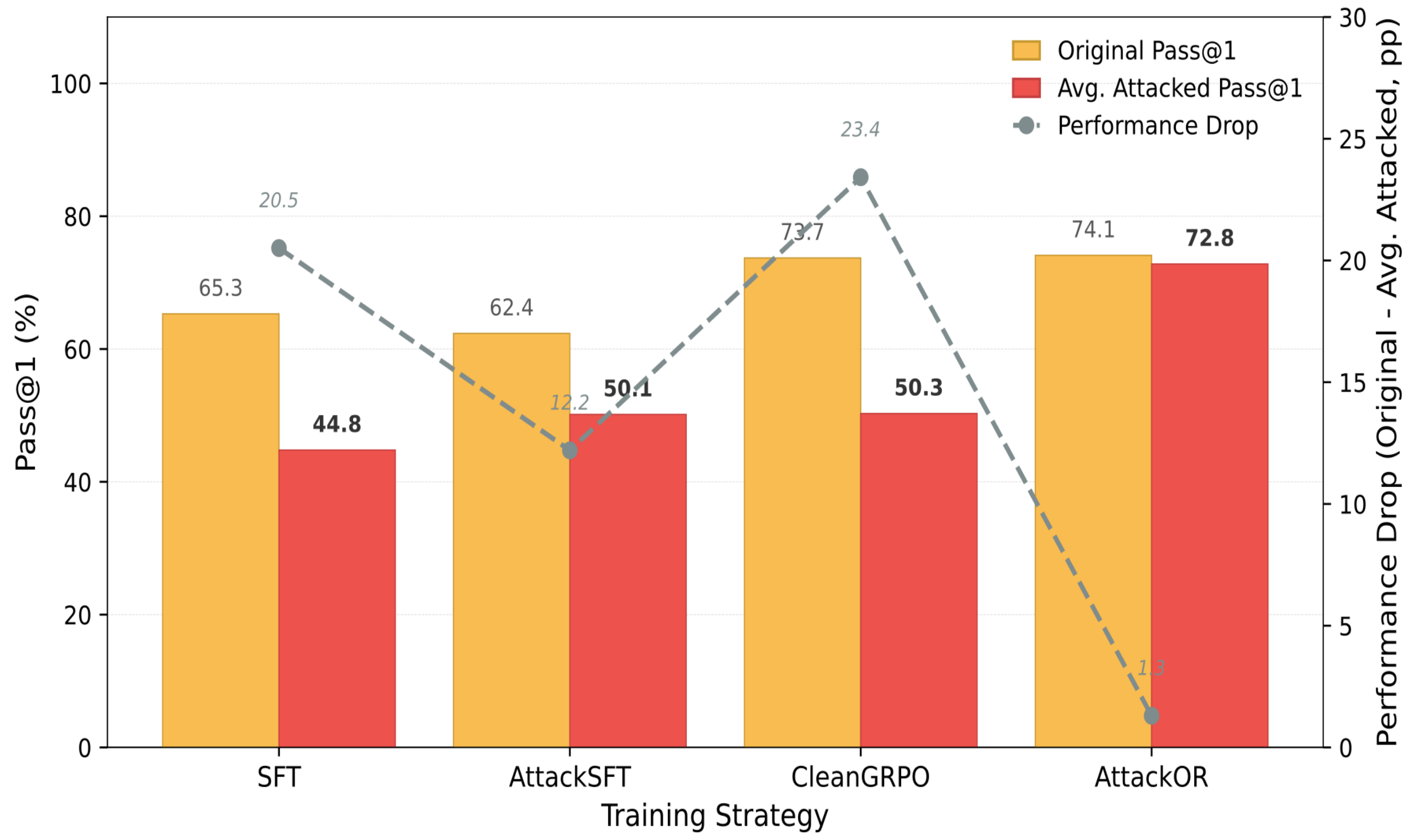
Main Results







Dataset	Model	Original	Attack family				Avg. Attacked
			Attribute	Phantom	Arithmetic	Unit	
NL4OPT	SFT	90.91	60.00	53.12	58.82	80.00	62.99
	ATTACKOR	**95.45**	**88.00**	**90.62**	**82.35**	**92.00**	**88.24**
IndustryOR	SFT	33.33	23.08	25.00	38.46	16.67	25.80
	ATTACKOR	**44.44**	**46.15**	25.00	**53.85**	**50.00**	**43.75**
MAMO Easy	SFT	86.96	50.79	56.06	84.91	54.29	61.51
	ATTACKOR	**89.86**	**87.30**	**83.33**	**88.68**	**92.86**	**88.04**
MAMO Complex	SFT	50.00	44.44	16.67	41.18	13.04	28.83
	ATTACKOR	**66.67**	**55.56**	**58.33**	**88.24**	**82.61**	**71.19**
Avg.	SFT	65.30	44.58	37.74	55.84	41.00	44.79
	ATTACKOR	**74.11**	**69.25**	**64.32**	**78.28**	**79.37**	**72.81**



(b) Attack-family generalization.

Model	Seen	Unseen	All
SFT	41.11	55.84	48.89
ATTACKOR	70.05	75.34	73.06
ATTACKOR-Subset	70.71	70.70	71.46

(a) Dataset-level generalization.

Model	Seen	Unseen	All
SFT	39.21	61.51	44.78
ATTACKOR	67.73	88.04	72.81
ATTACKOR-Subset	56.21	78.58	61.80

Model	Avg. Pass@1	Exec. Success	Mean Rel. Gap
Qwen3.5-9B-ATTACKOR	**73.07**	**98.99**	24.69
w/o R_{def}	64.05	98.40	62.88
w/o Analyzer	62.51	97.84	**11.14**